Additive Manufacturing of Aluminum Using Friction Stir Deposition

Ebtessam Elfishawy, M. M. Z. Ahmed and M. M. El-Sayed Seleman

Abstract In the current work, additive manufacturing process using friction stir deposition technique from die-cast Al–Si bar of 20 mm diameter against aluminum substrate was carried out. The effect of the spindle rotation speed (1200 rpm) and the feeding speed (3–5 mm/min) on the deposition process, the microstructure, and the hardness of the friction deposited parts were investigated. During the additive manufacturing process, after fixing the bar on the spindle shank and while rotating, the bar approaches the substrate at a constant feeding speed, and the bar plastically deformed due to the friction between the rotating bar and the fixed substrate under the effect of axial loading that causes the material to transfer from the bar to the substrate under the continuous feeding and the severe plastic deformation. This has resulted in a successful additive manufacturing of three cylindrical parts of 25 mm diameter of different heights at the different feeding rates. The additive manufactured parts (AMPs) were found to have a sound structure with ultra-fine grains.

Keywords Additive manufacturing · Friction stir deposition · Friction stir welding · Friction stir processing · Aluminum alloys · Hardness

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The Minerals, Metals & Materials Society (ed.), *TMS 2020 149th Annual Meeting & Exhibition Supplemental Proceedings*, The Minerals, Metals & Materials Series, https://doi.org/10.1007/978-3-030-36296-6_21

Introduction

The industrial history can be classified into periods separated by four industrial revolutions; the first revolution starts with the transition from manual production to steam-driven manufacturing between 1760 and 1830, the second technical revolution was ignited by the development of large scale material and energy production between 1840 and 1940, the third digital-driven revolution was initiated by transferring from analog, mechanical, and electronic systems to highly connected digital technology which starts from 1945 up to now, and the fourth industrial revolution which is known as direct digital manufacturing starts since 2010 which was driven by embracing much contemporary automation, data interconnectivity, and advanced manufacturing technologies [1]. In the fourth industrial revolution, additive manufacturing (AM) is considered as the key component for technological applications [2, 3]. Fusion beam-based AM methods are the most techniques used for metal-based AM [4]. However, achieving high structural performance is a main challenge because the liquid–solid phase transformation that makes their structural properties is limited by solidification microstructures[5]. Moreover, Anisotropic mechanical properties and high cost of powder manufacturing and preparation are another challenges that increase the gap between the beam-based AM capabilities and the industrial demand. That's why, solid-state AM technique has been introduced as a promising application of non-beam-based AM to bridge these gaps [2, 6]. As per the American Society for Testing and Materials (ASTM) classification, there are seven techniques of additive manufacturing. But the main approaches for metals and alloys manufacturing are the beam-based technologies such as powder bed fusion and directed energy deposition [1]. Those techniques are considered as high-energy and high-temperature processes which depend on melting the metal powder by laser or electron beam [7, 8]. As result of the rapid epitaxial solidification of the metal, the microstructure of the final product is highly orientated with columnar grains and anisotropic mechanical properties [1, 2, 7].

As most applications require isotropic mechanical properties, the non-beam-based solid-state additive processes have been developed to provide the isotropic structure required and to avoid the other limitations of beam-based additive manufacturing like the high-cost and time-consuming process of preparing suitable powders, in addition to the difficulty of fabricating non-weldable alloys, such as 2xxx or 7xxx Al alloys [2]. Afterwards, solid-state additive manufacturing has lagged behind in the form of friction stir additive manufacturing depending on the main concept of friction stir welding and friction stir processing [9, 10].

In 2011, Dilip et al. [11] succeeded to build a solid cylinder of 20 mm dia and 50 mm height with more than 30 layers by friction deposition using a 19-mm-diameter consumable rod of austenitic stainless steel AISI 304 against a mild steel substrate of dia (25 mm) and the following parameters: 800 rpm spindle rotation speed, 9.8 kN ton friction force, and 3 mm burn-off were used [12]. Non-beam-based additive manufacturing has many techniques; one of them is the seam welding of multi-layered similar and dissimilar metallic sheets which has been introduced by Prasad et al. in

2015. In their study, a defect-free, metallurgically bonded, multi-track, and multi-metal seam welds were obtained of relatively high-temperature melting materials, AISI 304, C-Mn steels, Ni-based alloys, CP Cu, CPNi, Ti6Al4V, and relatively low-temperature melting material, AA6061. This process depends on the rotation of non-consumable rod (CP Mo or AISI 304) against the upper sheet of a clamped metallic stack [13]. Then, an axial force was applied once the distal end of the rod touched the top portion of the upper metallic sheet. And to get the desired welding length, the metallic stack moved horizontally relative to the stationery non-consumable rod after an initial dwell time causing the metallurgical bonding between the metallic sheets which was attributed to localized stick and slip at the interface, dynamic recrystallization, and diffusion. The mechanical tensile shear testing showed that the seam welds were generally stronger than the base metal counterparts [13]. In 2015, using friction deposition, Karthik et al. [14] succeeded to make a metal–metal composite of aluminum matrix with titanium particle reinforcement (6 vol.% Tip/AA5083). The friction deposited multi-layer composite microstructure was very fine grain size including uniformly distributed titanium particles without any harmful intermetallic formation at the reinforcement/matrix interfaces. The produced layers were well-bonded. [14].

Another technique of additive manufacturing was developed in 2017 by Bandari et al. [15], who have worked on commercial pure aluminum with a thickness of 6 mm as the base plate and TiC particles of average size 44 m which were used as reinforcement to form tailor-made functionally graded composites by friction stirring. In addition to control reinforcement particle volume fraction using a mathematical model by managing their distribution in a matrix and counting on the ability of friction stirring for local homogenization that leads to property gradients, Sharma et al. [15] also concluded that the parameters which affect mixing of particles and grain size of matrix are the particle size [16, 17], volume fraction, number of stirring passes, and change or continuation of tool traverse direction in successive passes. While those that affect the mechanical properties are continuous dynamic recrystallization by helping grains to grow, crushing of reinforcement particles restricts grain boundary migration and crushing of the initial matrix grain. Hence, changes in local properties can be controlled by achieving control over the volume fraction. Therefore, the observed defects after initial pass of stirring like cracks and voids can be eliminated after multiple passes. This developed approach results in gradients in mechanical properties such as yield stress, Young's modulus, and the strain hardening coefficient which can be evaluated with the digital image correlation technique [15].

Recently, Rivera et al. [18] have succeeded in 2019 to deposit AA2219 material at high deposition rate (1000 cm^3/h) with the aid of solid-state additive deposition process depending on additive friction stir deposition or MELD. MELD technique significantly differs from additive friction stir deposition where now material is extruded through a hollow rotating tool [11]. This process allows for repairing, coating, and/or building fully dense materials, as it is a highly scalable process, with high deposition rates for aluminum alloys over 1000 cm^3/h. Rivera et al. [18] studied the influence of texture and grain refinement on the mechanical behavior of AA2219; they found

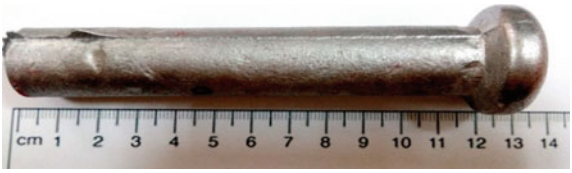

Fig. 1 Consumable rod (20 mm diameter, as -received die-cast Al–Si

an equiaxed grain morphology in the three orientations. Moreover, while using electron backscatter diffraction (EBSD), they identified a layer-dependent texture in the top of the build with a strong torsional A fiber texture which was transitioning to weaker textures in the middle and bottom layers. Consequently, the tensile strength increases from the bottom to the top of the deposition which reflects the texture layer dependence. However, the statistically measured hardness showed no significant differences from the top to the bottom of the deposition. Furthermore, mechanical properties were observed to have no orientation dependence for both compression and tension specimens tested. Transmission electron microscopy (TEM) illustrated a lack of θ' precipitates in the as-deposited cross section, which is resulting in no precipitation strengthening [18]. The current study investigates the use of Al–Si cast rod as the consumable rod to produce FSAMed simple parts. The effect of the rotation rate and feed speed on the microstructure evolution and hardness will be investigated.

Experimental Work

Material

Consumable rods (picture shown in Fig. 1) of die cast Al–Si of 20 mm diameter and 100 mm length have been used as the starting material in the friction stir additive manufacturing process.

Friction Stir Additive Manufacturing

Friction stir additive manufacturing was conducted using homemade FSW machine [9]. The difference between FSW&FSP [19] and FSAM is that the used tool in FSW&FSP [20] is non-consumable, hard, and wear resistant tool, unlike the soft consumable tool to be used in FSAM. Most of the previous studies use a continuous feeding extruded wire through a hollow non-consumable tool. This study is to investigate the mechanism of solid-state friction deposition in the vertical alignment using a consumable tool to make the additively manufactured parts (AMPs). Figure 2 shows the friction stir welding machine at Suez University that used to perform the friction stir deposition.

Fig. 2 FSW machine at Suez University

Different parameters were used to investigate the effect of friction stir deposition process parameters on the microstructure of the produced parts. After careful optimization of process parameters, the following parameters were investigated: 1200 rpm spindle rotation speed with 3, 4 and 5 mm/min feeding speeds was examined using consumable rod of die-cast Al–Si. All experiments were done at 0° tilt angle.

Characterization

The additively manufactured parts (AMP) have been sectioned vertically along the building direction and the longitudinal sections were prepared according to the standard metallographic procedures of grinding and up to mechanical polishing using 0.3 μm alumina suspension and then investigated using the optical microscopy after etching using Keller's etchant of chemical composition of 25 ml methanol, 25 ml hydrochloric acid, 25 ml nitric acid, and 1 drop hydrofluoric acid.

Results and Discussion

Macro-observations

During the process after fixing the consumable bar on the spindle shank and while rotating, the consumable bar is exposed to severe plastic deformation and heat-up due to the friction between the rod and the substrate. That causes the consumable

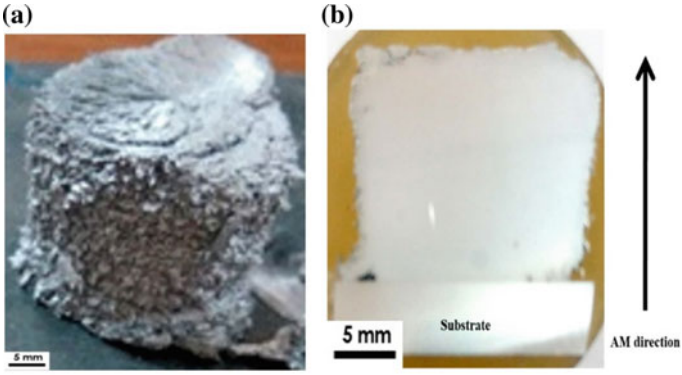

Fig. 3 Cylindrical part produced by FSAM at feeding rate of 3 mm/min, and spindle rotational speed of 1200 rpm. **a** As additively manufactured part, **b** longitudinal section of the FSAMed part

rod tool to soften and deposit on the substrate in a continuous manner as long as the feeding continues. The material continues to transfer from the consumable rod to the substrate in the form of well-bonded layer upon layer to finally form an additively manufactured cylinder by solid-state friction deposition. For more understanding of the process, the material of the tool under hydrostatic pressure is plastically deformed and deposited on the substrate as they were stirred together to metallurgically bond the material to the substrate and the successive layers together as well. This causes the success of producing the AM cylindrical parts with dimensions' range about 20–35 mm height and 25 mm dia which was built by friction stirring deposition as shown in Fig. 3a. The friction stir additive manufactured (FSAMed) parts were completely sound and continuous without any separations between the internal layers along the AM building direction as shown in Fig. 3b so that the number of layers could not be measured internally.

A typical section of the AM parts is shown in Fig. 4 which clearly illustrates the continuity of the process. Moreover, the macrograph assures the sound defect-free structure of the AM parts.

The Consumable Tool Interface

The investigation of the used consumable tool microstructure showed that at the tool tip adjacent to the substrate, the processing surface of the tool has a refined recrystallized grains due to severe plastic deformation. The grain size increases gradually going up to reach the base metal coarse grain structure as shown in Fig. 5. This microstructure shows the effect of friction stirring process on refining and recrystallizing the grains and the silicon precipitates. In Fig. 5, on the right hand a macrograph of the tool side section is shown where the processing side is down which faced the substrate and was exposed to the friction. On the left hand, the different structure of

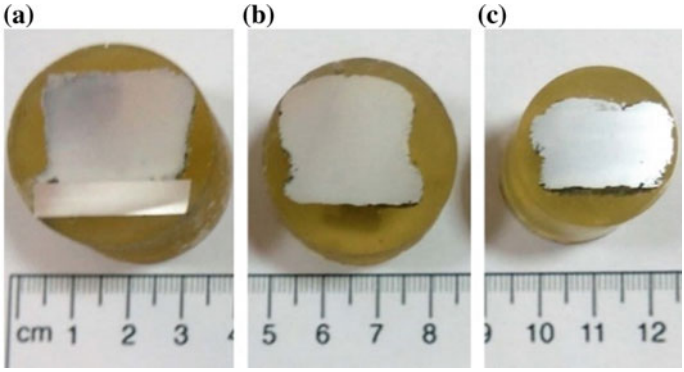

Fig. 4 Optical macrographs of the FSAMed parts (longitudinal section) at spindle rotational speed of 1200 rpm at feeding rates of **a**3 mm/min, **b** 4 mm/min and **c** 5 mm/min

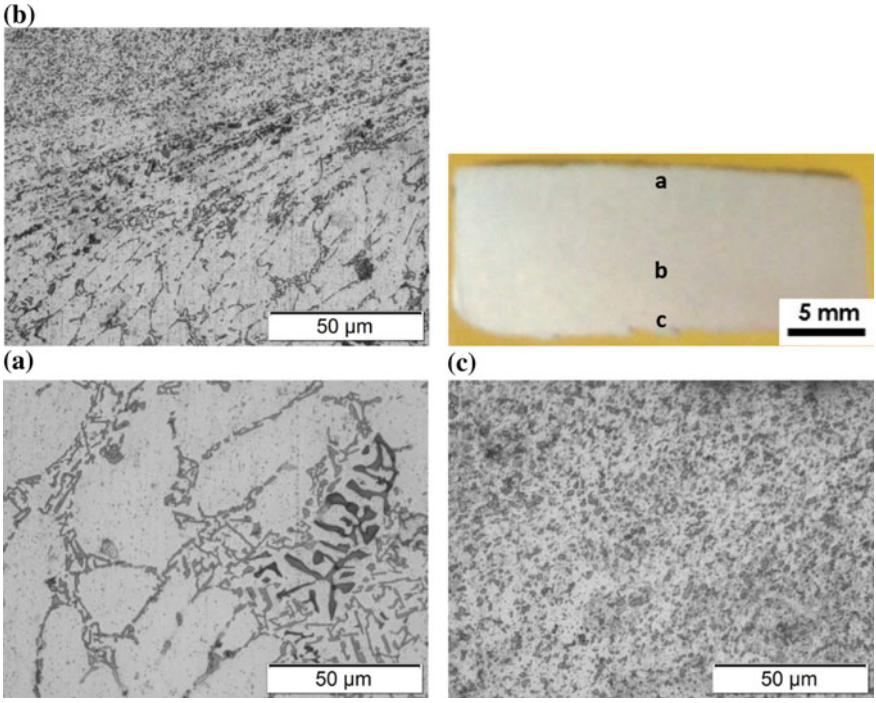

Fig. 5 Macro- and micrograph of the used consumable rod (longitudinal section) at feeding speed of 4 mm/min and spindle rotational speed of 1200 rpm **a** microstructure of the above area of the tool near the base microstructure, **b** the transition zone, **c** the recrystallized fine structure which was exposed to friction with the substrate

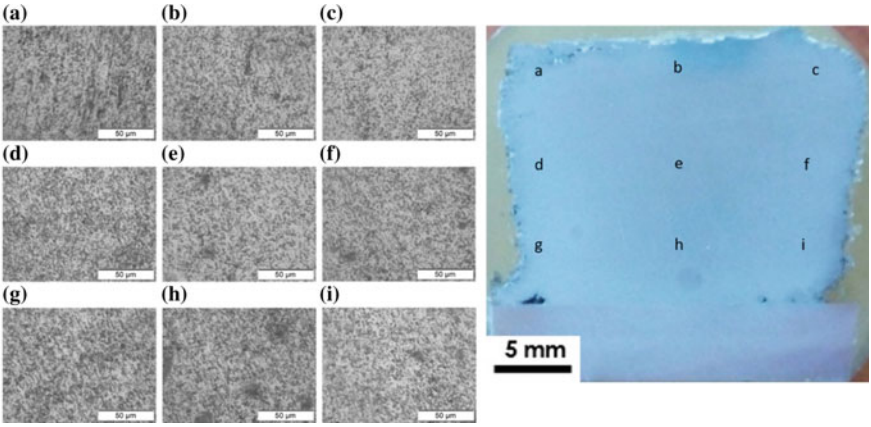

Fig. 6 Macro- and microstructure of Al–Si AMP at feeding rate of 3 mm/min, and spindle rotational speed of 1200 rpm

the tool illustrating the effect of the friction on refining the tool tip grains as shown in (c), going up through the tool, the transition zone (b) between the fine structure of the recrystallized tool tip and the base coarse grain structure (a) is shown. The consumable die-cast rod alloy Al–Si has a relatively coarse grain structure as shown in Fig. 5a, where the silicon is forming the coarse grain boundaries. During friction stirring deposition, the silicon precipitates on the grain boundaries were broke into fine particles and redistributed in the plastically deforming aluminum matrix (Fig. 6c). Thermal treatment with severe plastic deformation causes silicon to coalesce and spheroidize.

Additive Manufactured Parts (AMPs) Microstructure

Microstructural examination of the AMPs was carried out using an Olympus upward Microscope. The optical microscopy examination was carried out on long-transverse sections of the cylindrical AMPs. The microstructure investigation of the AMPs showed very fine grain structure as that is produced in the nugget zone of FSW. That's why the stirring zone in the FSAM follows the same dynamic recrystallization (DRX) [13, 21]. There are two types of DRX, continuous DRX and discontinuous DRX. The difference between them is that new grains are formed by a gradual increase in the misorientation between the subgrains in case of the continuous DRX. In the case of discontinuous DRX, grains with high-angle grain boundaries form through dynamic nucleation and growth from a previously deformed microstructure [22]. Previous research works have concluded that aluminum alloys commonly undergo dynamic recovery (DRV) during hot deformation [21]. Another study made by Su et al. on AA7075 has shown that Aluminum alloys in the FSW and FSP will experience

different mechanisms such as discontinuous DRX, DRV, and continuous DRX at different stages of the microstructural evolution [21, 23, 24]. This explained the formation of equiaxed fine grain morphology in the tip of the consumable tool and AMPs microstructure.

The microstructure investigation of the AMPs has been studied in different positions along the specimen. This helps in better understanding of the microstructure. In all feeding rates as shown in Figs. 6, 7, and 8, the microstructure is homogeneous and nearly the same in any position of the specimen. This fine equiaxed microstructure is the aim which most of AM researchers are seeking for, to fulfill the industrial requirements of isotropic mechanical properties. The micrographic study also shows the refined grains and the spheroidal precipitates resulting from dynamic recrystallization.

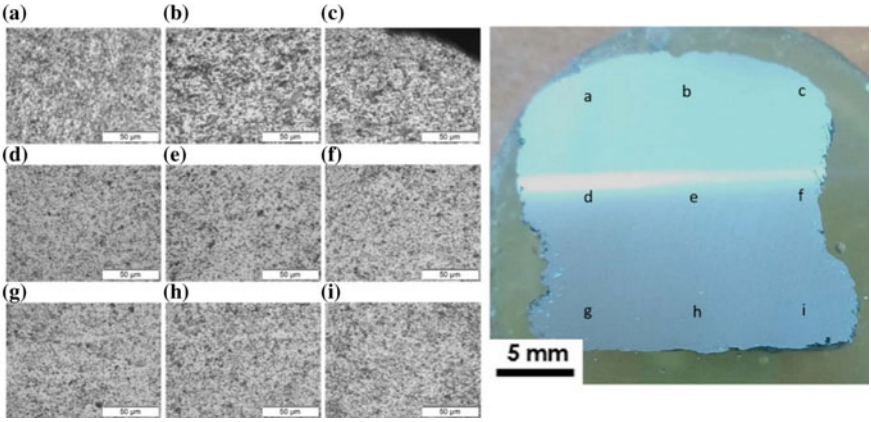

Fig. 7 Macro- and microstructure of Al–Si AMP at feeding rate of 4 mm/min, and spindle rotational speed of 1200 rpm

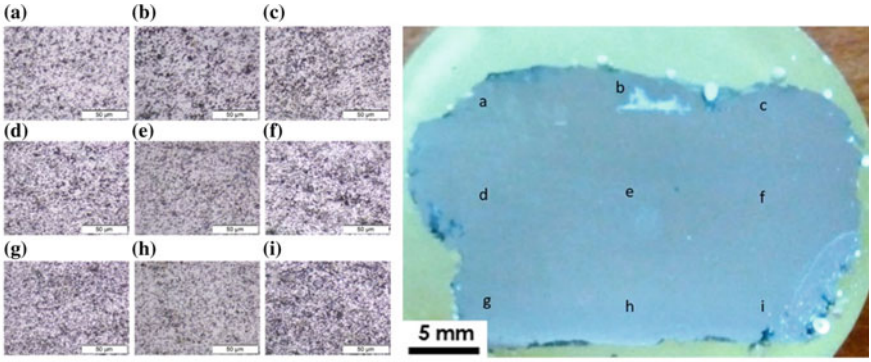

Fig. 8 Macro- and microstructure of Al–Si AMP at feeding rate of 5 mm/min, and spindle rotational speed of 1200 rpm

Table 1 Hardness values of the base Al–Si rod and the AMPs

Specimen no.	Rotation speed (rpm)	Feeding speed (mm/min)	Average hardness (HV)
Base			85.86
1	1200	3	81.08
2	1200	4	79.84
3	1200	5	82.14

Hardness

Bulk hardness measurements were carried on the longitudinal sections of the cylindrical AMPs using a Vickers hardness tester. The results in Table 1 show the average of 5 readings which were taken on each sample. There were no statistically significant differences in hardness measured from the top to the bottom of the deposition. The hardness results showed no significant change of the hardness properties of the material before and after FSAM. This implies that the grain refining has almost no effect on the hardness and also the fragmented stable particles also have not affected the hardness of the AMPs.

SEM Analysis

SEM examination of the AMPs revealed significant fragmentations for the intermetallic particles that were existing in coarse sizes in the base material. This reflects the effect of the stirring process in the fragmentation and distribution of the stable intermetallic particles (Fig. 9).

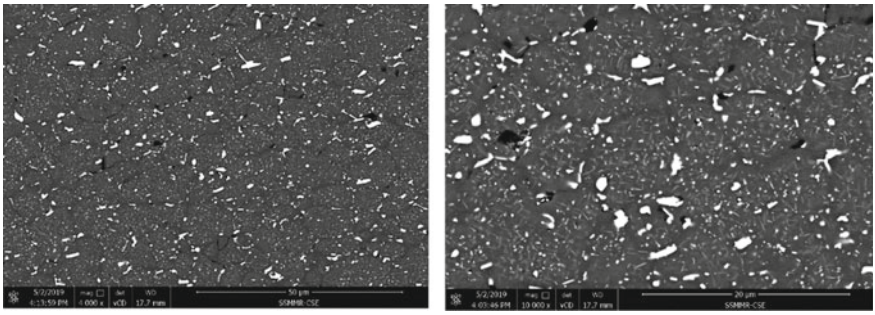

Fig. 9 SEM micrographs for the Al–Si AMPs using 1200 rpm and 6 mm/min feeding speed at different magnifications

Conclusion

This study investigates the microstructure and mechanical properties of AMPs produced using die-cast Al–Si consumable rod by continuous friction stir deposition process, and the following conclusions can be drawn:

- Al–Si AMPs were successfully produced by friction stir deposition in multiple layers of thickness up to 30 mm without any physical discontinuities or interfacial defects between layers. Moreover, the parts are homogenous and continuous without any separation between the layers.
- The current investigation establishes the feasibility of a novel additive manufacturing technique depending on the high-performance continuous friction stirring process.
- The Al–Si AMPs produced in the current work feature several impressive microstructural characteristics such as dynamically recrystallized, ultra-fine grains with very fine, well-distributed precipitates, and homogeneous and equiaxed microstructure.
- The grain structure formed in the AMP is significantly fine grain structure ranging between 2 and 5 μm relative to the base material grain structure that was coarse grain structure of about 50 μm. In addition, the coarse intermetallic in the cast alloy has been significantly refined and uniformly distributed in the matrix alloy.

Acknowledgements Science and Technology Development Fund is acknowledged for funding this work with grants no. 3926 and 5304.

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